

D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
X	X	X	X	ADDRESS				MSB		DATA				LSB	

Fig. 3: Serial data format

coded decimal decoder, multiplex scan circuitry, segment and digit drivers, and

static RAM that stores each digit-dis- play information

play information stores each digit-dis- play information. Resistor R5 limits the current through segment LEDs.

Each and every function of MAX7219 is addressable and programmable by memory locations (refer Table I). The data is transferred to MAX7219 in serial format as shown in Fig. 3.

Serial data at D_{IN} is converted into 16-bit format and shifted into the internal 16-bit shift register with each rising edge of CLK. LOAD pin is held low during this process and must go high concurrently after sixteenth rising clock edge.

Table I shows the 14 addressable digit and control registers of MAX7219. Address locations 0xX1 through 0xX8 are used to show the data on dot-matrix LED display DIS1. Data to be displayed is controlled by the decode-mode register, which is available at address 0xX9. Display brightness can also be controlled with resistor R5. The peak current sourced from the segment drivers is normally about

a hundred times the current entering at pin 18 (ISET) of MAX7219. Display brightness can also be controlled digitally by intensity register (address location 0xXA).

MAX7219 also offers the facility to shut down the display. Normal operations can be resumed with the help of shutdown-mode register. This mode of operation is useful for power conservation.

To derive the power supply for the circuit, the 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 500 mA. The transformer's output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC 7805. Capacitor C2 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator and R1 limits the current through LED1.

An actual-size, single-side PCB for the microcontroller-based dice using dot-matrix display is shown in Fig. 4 and its component layout in Fig. 5. Assemble the circuit on a PCB as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error.

Software

The program is written in 'C' language and compiled using WinAVR. It is well-commented and easy to understand.

The software is used to:

1. Detect pressing of switch S2
2. Compute dice count
3. Control the various modes of MAX7219 operation
4. Control the flow of information to MAX7219 (generation of serial clock, conversion of 8-bit address and data into 16-bit serial data, etc)
5. Display the desired information on the LED matrix

The program files (refer Table II) used in this software are described below:

Dice.c. It is used to initialise AT-mega8's ports and detect the status of the key with the help of 'get_Key_Status()'

TABLE I
Register Address Map of MAX7219

Register	Address						Hex code
	D15-D12	D11	D10	D9	D8		
No-Op	X	0	0	0	0	0	0x00
Digit 0	X	0	0	0	1	0	0x01
Digit 1	X	0	0	0	1	0	0x02
Digit 2	X	0	0	0	1	1	0x03
Digit 3	X	0	0	1	0	0	0x04
Digit 4	X	0	0	1	0	1	0x05
Digit 5	X	0	0	1	1	0	0x06
Digit 6	X	0	0	1	1	1	0x07
Digit 7	X	1	0	0	0	0	0x08
Decode mode	X	1	0	0	0	1	0x09
Intensity	X	1	0	0	1	0	0x0A
Scan limit	X	1	0	0	1	1	0x0B
Shutdown	X	1	1	0	0	0	0x0C
Display test	X	1	1	1	1	1	0x0F

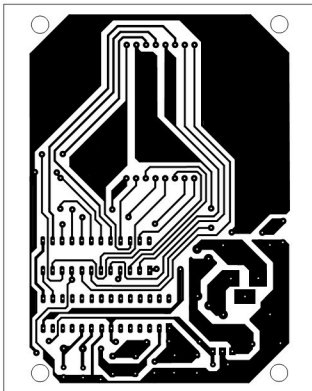


Fig. 4: An actual-size, single-side PCB for the microcontroller-based dice using dot-matrix display